Screens — Helix Console (admin web/ desktop app)

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> Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** the complete screen inventory and per-screen UX specification of **Helix Console** — the HelixVPN administration app (Flutter, `flavor: HelixFlavor.console`, capability `{Capability.admin}`, **no `helix_core_ffi`**, the **only** flavor that builds to Web) [`03-client-core-and-ui.md` §6, CI3]. It defines, screen by screen: purpose, desktop-first responsive layout, the `helix_design` components used, every state (loading / empty / error / partial / live), navigation, interactions, accessibility, and the **light + dark** rendering — each backed by an ASCII wireframe.
>
> **SPEC-ONLY.** It describes *what each screen is and how it behaves* — not the shipping `console` build. Original HelixVPN UX design work; the control-plane data each screen surfaces is owned by the cited control-plane docs.
>
> **Boundary with sibling docs. Owns:** the Console screen set + per-screen layout/state/interaction/a11y. **Consumes:** the colour roles + connection / feedback semantics + per-app **violet** accent [`color-system.md` §2/§6]; the token tiers + spacing/radius/breakpoint scales [`design-tokens.md` §6]; the signature components (`AdaptiveScaffold`, `NetworkTile`, `PolicyEditor`, …) [`component-library.md`, sibling, this wave]; the OpenDesign engine that emits the tokens [`opendesign-foundation.md`]; the **Connector** counterpart screens [`screens-connector.md`, sibling]; the **Client** screens [`screens-client.md`, sibling]. The **data** every screen renders is owned by the control-plane: [`02-control-plane.md`] + the V3 service docs [`v03-control-plane/svc-registry.md`], [`…/svc-policy.md`], [`…/svc-coordinator.md`], [`…/svc-telemetry.md`], [`…/svc-ipam.md`], [`…/svc-identity.md`].
>
> **Evidence base.** `[CP §N]` = `final/02-control-plane.md`; `[CLIENT §N]` = `final/03-client-core-and-ui.md`; `[REG §N]`/`[POL §N]`/`[COORD §N]`/ `[TEL §N]`/ `[IPAM §N]` = the matching `final/v03-control-plane/svc-*.md`; `[COLOR §N]` = `final/v10-design/color-system.md`; `[DT §N]` = `final/v10-design/design-tokens.md`. Claims not grounded in the

> evidence base or in this document's own original design choices are tagged `UNVERIFIED` per constitution §11.4.6 — never fabricated. Layout/state/interaction designs are **original HelixVPN UX work**.

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0. What the Console is — and the design principles that shape every screen

Helix Console is the **admin brain's front end**: it is where a tenant administrator manages *who may reach what*. It is the UI over the Go control plane's REST + WebSocket/SSE surface [CP §8] — it **never** opens a tunnel, never links the Rust core (`Capability.admin`, no `helix_core_ffi`, CI3 [CLIENT §6]) — so it is the one flavor that builds to **Web** and also ships as a desktop app (macOS / Windows / Linux from the same tree).

Five principles govern every Console screen:

1. **Density without clutter — desktop-first.** The Console is a power tool used on `lg` (1024) and `xl` (1440) viewports [DT §6.6]; it presents heavy tables, a graph, and a code-like policy editor. It uses the **expanded `AdaptiveScaffold`** (extended `NavigationRail` + multi-pane) [CLIENT §7.3] and degrades gracefully to a phone layout, but the home turf is the wide screen.
2. **Authority accent = violet, safety colours unchanged.** The Console's brand-overridable accent is `violet.*` [COLOR §6] — it reads as an admin/authority surface, distinct from the Client's indigo and the Connector's teal. The **connection-state and feedback semantics never change** (D-COLOR-1): a device shown "online/protected" is the same green everywhere.
3. **Live, not polled.** Device lists, the topology graph, and the audit feed fold `GET /v1/stream` WS/SSE events (`device.online`, `route.changed`, `policy.compiled`, `device.revoked`) into state and update **without a manual**

refresh [CLIENT §8.3, CP §8]. A small live-connection indicator tells the admin the stream is healthy.
4. **Fail-closed, preview-before-apply.** The highest-stakes screen (Policy editor) compiles **dry-run first** and shows an effect-diff before anything activates [POL §4/§5]; destructive control actions (revoke, deactivate) require explicit confirmation per §11.4.66.
5. **Privacy by construction is visible.** The Console **cannot** show traffic / connection / flow logs because they do not exist (C3 [CP §0.1]). The Audit screen states this plainly — it shows *control actions*, never user traffic.

> **Honest boundary (§11.4.6).** This document specifies layout, state, and interaction. It does **not** re-specify the REST/WS payloads (owned by [CP §8] + the V3 service docs) nor the token hex/contrast (owned by [COLOR]/[DT]). Where a concrete control-plane field name is cited it is grounded in those docs; a value not so grounded is marked `UNVERIFIED`.

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1. The Console shell (AdaptiveScaffold, responsive law, command palette)

Every screen renders **inside** the Console shell. The shell is `AdaptiveScaffold` [CLIENT §7.2] branching on **width, never `Platform.isX`** [CLIENT §7.3]:

Breakpoint [DT §6.6]	Width	Shell layout
`compact`	< 600	`BottomNavigationBar` (5 primary destinations) + single pane; overflow destinations in a "More" sheet
`medium`	600–1024	`NavigationRail` (icons + labels) + master/detail two-pane
`expanded`	> 1024	**Extended `NavigationRail`** (icon + text, pinned) + multi-pane (list │ detail │ live-events rail) — the Console home turf

```
┌──────────────────────────────────────────────────────────────────
│ Helix Console      [acme-corp ▾]           ⌘K  ◐ theme   ◉
live   ⦿ admin │  ← top app bar (surface.default, violet brand
mark)
├────────────────┬─────────────────────────────────────────┬───────
│ ▣ Dashboard    │                                         │  LIVE
EVENTS  │
│ ▤ Devices      │          (primary content pane)          │
───────────   │
│ ⦿ Users & SSO │                                         │
12:04 dev on  │
│ ⛨ Policy       │                                          │ 12:03
policy  │
│ ◈ Topology     │                                         │
v8 act.  │
│ ▦ IPAM         │                                         │ 12:01
dev off │
│ ≡ Audit        │                                          │
```

```
…              │
│ ▲ Telemetry    │                                           │
(folds /v1/    │
│ ▮ Billing      │                                           │
stream WS)    │
│ ⚙ Settings     │
│                │
│ ─────────────  │
│                │
│ ◉ stream OK    │
│                │
└────────────────┴───────────────────────────────────────────┴───────
   extended rail              content (master│detail)
live-events rail
```

- **Top app bar** carries: the Console brand mark (violet), the **tenant switcher** `[acme-corp ▾]` (an admin with rights to >1 tenant; one tenant ⇒ static label), the ⌘K **command palette** trigger, a **theme toggle** (◐ system/light/dark), the **live-stream indicator** (◉ green = stream healthy / amber = reconnecting / red = stream down — reuses the connection-state palette [COLOR §3]), and the signed-in admin avatar + role chip.
- **Command palette (⌘K / Ctrl-K)** [CLIENT §7.3] — fuzzy jump to any screen, device, user, policy version, or action ("revoke device …", "activate policy v7"). Keyboard-first; every action it lists is also reachable by mouse.
- **Live-events rail** (expanded only) folds the WS/SSE stream; on `medium`/`compact` it collapses into a bell/notification sheet.
- The **stream-health chip** at the rail foot is the §11.4.6-honest signal that "live" is real: amber when the WS reconnects, red + a banner when it is down (the lists then show a "may be stale — reconnecting" notice rather than lying).

---

2. Screen inventory + navigation map

#	Screen	Route	Primary data source	Min role [CP §8.1]	Key components
S0	Sign-in / tenant select	`/signin`	identity (OIDC) [`svc-identity.md`]	(unauth)	brand splash, OIDC button, tenant chooser
S1	Dashboard / overview	`/`	aggregate of registry + telemetry + policy	member+	stat cards, SLO sparkline, live-event digest
S2	Devices — list	`/devices`	`GET /v1/devices` [REG §3]	member+	data table, presence dot, filter bar
S2d	Device — detail	`/devices/:id`	device + certs + presence [REG §2/§5]	member+	detail header, cert panel, revoke action
S3	Users & SSO	`/users`		admin	roster table, role editor,

#	Screen	Route	Primary data source	Min role [CP §8.1]	Key components
			identity / OIDC config [`svc-identity.md`]		OIDC config form
S4	Policy editor	`/policy`	`policies.spec` + compiler [POL §2/§4]	operator+	`PolicyEditor`, effect-diff, version history
S5	Topology	`/topology`	coordinator graph [COORD §1]	member+	graph canvas, node/edge inspector
S6	IPAM / prefixes	`/ipam`	overlay pool + advertised_prefixes [IPAM §1, REG §6]	operator+	pool card, prefix table, conflict list
S7	Audit	`/audit`	`audit_events` [TEL §4]	admin (read)	audit feed, action filter, no-traffic banner
S8	Telemetry / health	`/health`	Prometheus series [TEL §3]	operator+	SLO gauges, convergence histogram, alert list
S9	Billing / subscription	`/billing`	billing provider `UNVERIFIED`	admin	plan card, seat usage, invoices
S10	Settings	`/settings`	tenant config + per-user prefs	varies	tenant settings, theme, session, danger zone

```
flowchart TD
    S0["S0 Sign-in / tenant select"] -->|authenticated| S1["S1 Dashboard"]
    S1 --> S2["S2 Devices (list)"]
    S1 --> S4["S4 Policy editor"]
    S1 --> S5["S5 Topology"]
    S1 --> S8["S8 Telemetry / health"]
    S2 --> S2d["S2d Device detail"]
    S2d -->|revoke| S2
    S1 --> S3["S3 Users & SSO"]
    S1 --> S6["S6 IPAM / prefixes"]
    S1 --> S7["S7 Audit"]
    S1 --> S9["S9 Billing"]
    S1 --> S10["S10 Settings"]
    S4 -. "compile error → device/host refs" .-> S2
    S4 -. "host CIDR not advertised → fix" .-> S6
    S5 -. "click connector node" .-> S6
    S5 -. "click device node" .-> S2d
    S7 -. "policy.activate row → version" .-> S4
    CMD["⌘K command palette"] -. "jump to any screen / entity / action" .-> S1 & S2 & S4 & S5 & S7
```

The map is a hub-and-spoke from the Dashboard, with **cross-links that follow the data**: a policy compile error that names an un-advertised host CIDR links to IPAM (S6); a topology connector node links to its IPAM prefixes; an audit `policy.activate` row links to that policy version in the editor.

---

3. Sign-in (OIDC + tenant select)

Purpose. Authenticate a tenant admin (OIDC session) and, when the identity maps to more than one tenant, choose which to administer. Managed (OIDC) is the Console's identity mode [CP §9.1] — the anonymous device-token path is a *Client* enrollment concern, not a Console sign-in.

Layout (desktop-first, centred card).

```
┌──────────────────────────────────────────────────────────────────
│
│
│                      ◆  Helix
Console                                         │  ← violet brand
mark
│                      Administer your private
network                       │
│
│
│
┌─────────────────────────────────────────────┐                │
│              │  Continue with your identity provider
│                │
│              │  ┌──────────────────────────────────────┐
│                │
│              │  │   ☉  Sign in with SSO  (OIDC)         │  │
← action.primary (violet)
│              │  └──────────────────────────────────────┘
│                │
│              │
│                │
│              │  Self-hosted? Enter your control-plane URL
│                │
│              │  ┌──────────────────────────────────────┐
│                │
│              │  │ https://cp.acme.example              │  │  ←
input
│              │  └──────────────────────────────────────┘
│                │
│
└─────────────────────────────────────────────┘                │
│
│
│      No traffic is ever logged. Helix stores identity & policy
only.       │  ← privacy note (C3)
└──────────────────────────────────────────────────────────────────
```

Tenant select (post-auth, multi-tenant identity).

```
┌───────────────────────────────┐
│  Choose a tenant              │
│  ┌─────────────────────────┐  │
│  │ ◈ acme-corp     142 dev │  │ ← selectable rows; presence
count from registry
│  │ ◈ acme-eu        38 dev │  │
│  │ ◈ lab            6 dev   │  │
│  └─────────────────────────┘  │
└───────────────────────────────┘
```

States. *Loading* — brand splash + indeterminate progress while the OIDC redirect resolves. *Empty* — identity maps to **zero** tenants ⇒ a clear "your account is not an admin on any tenant; contact your operator" message (not a spinner that hangs). *Error* — OIDC failure shows the honest provider reason (`feedback.error`, [COLOR §2.4]) with a retry, never a generic "something went wrong"; an unreachable control-plane URL is a distinct, actionable error.

Interactions / a11y. SSO button is the default focus + Enter-activates; the control-plane-URL field validates as a URL on blur. Full keyboard nav; the brand mark has an `alt`. **Light/dark:** the card is `surface.raised`, the SSO button `action.primary` (violet `#6D28D9` light / `#A78BFA` dark, both AA-proven [COLOR §6]); the privacy note is `text.secondary`.

---

4. Dashboard / overview

Purpose. The at-a-glance health of the tenant's network: how many devices are online, whether the active policy is healthy, whether the control plane is meeting its < 1 s convergence SLO [CP §10.2], and a digest of recent control activity. It is a **read** screen (member+) that links into every deeper screen.

Layout (expanded, three-zone).

```
┌──────────────────────────────────────────────────────────────────┐
│  Dashboard — acme-corp                                   updated
live ◉    │
├──────────────────────────────────────────────────────────────────
│  ┌───────────┐ ┌───────────┐ ┌───────────┐ 
┌───────────┐                               │
│  │ DEVICES   │ │ CONNECTORS│ │ ACTIVE    │ │ CONVERGE  │   ←
stat cards    │
│  │ 142       │ │ 7         │ │ POLICY    │ │ p99       │
(elevation.card)│
│  │ 118 online│ │ 6 online  │ │ v8 ✓      │ │ 0.42 s ✓
│              │
│  │ ▲ green   │ │ ▲ green   │ │ green     │ │ green
│              │
│  └───────────┘ └───────────┘ └───────────┘
└───────────┘                                 │
│  ┌───────────────────────────────────┐
┌───────────────────────────────┐    │
```

```
│  │ CONVERGENCE (event→delta, p99)    │  │ RECENT CONTROL
ACTIONS        │   │
│  │  1s┤                           │  │ 12:04 alice revoked
dev-31    │   │
│  │    │      ╭╮      ╭─╮          │  │ 12:03 system activated
pol v8 │   │
│  │ .5s┤ ╭──╯╱  ╲╭──╯  ╲─        │  │ 12:01 bob enrolled
connector  │   │
│  │    │╱          (sparkline)     │  │ 11:58 carol changed
prefixes  │   │
│  │  0 └──────────────────────────  │  │ → see full
Audit              │   │
│  └───────────────────────────────────┘
└───────────────────────────────┘    │
│
┌──────────────────────────────────────────────────────────────────
│
│  │ ⚠ 2 advisory route conflicts (overlapping CIDRs) — review in
IPAM      │ │ ← advisory banner (warning)
│
└──────────────────────────────────────────────────────────────────
│
└──────────────────────────────────────────────────────────────────
```

Components. Four **stat cards** (`elevation.semantic.card` [DT §6.3]) each with a value, a sub-count, and a state-coloured indicator dot; a **convergence sparkline** sourced from `helix_reconcile_seconds` p99 [TEL §3.2, COORD §7.3]; a **recent-actions digest** (last N `audit_events` [TEL §4]); an **advisory banner** summarising open `route.conflict.detected` items [REG §6.3] (advisory, never blocking — links to IPAM S6).

States. *Loading* — skeleton cards (shimmer placeholders, never a blank screen). *Empty* — a freshly-bootstrapped tenant with 0 devices shows an onboarding card: "Enroll your first connector" → mint an enroll token (links to Devices/enroll). *Error* — if an aggregate query fails, that one card shows an inline error + retry while the rest render (no all-or-nothing). *Live* — counts and digest update from `/v1/stream` without refresh; the "updated live ◉" chip goes amber if the stream drops.

SLO colour logic (honest, §11.4.6). The convergence card is green only when p99 < the SLO target (1 s [CP §10.2]); amber within 1–2×; red above — and the card states the *measured* number, never "healthy" without the figure.

A11y / light-dark. Stat values are `text.primary` (AAA both themes); the indicator dots carry a text/icon label too (never colour-only, CI5 [CLIENT §0.1]). Cards are `surface.raised`; the warning banner is `feedback.warning` (`amber.700` light / `amber.400` dark [COLOR §2.4]).

---

5. Devices (list · detail · approve / revoke)

5.1 Devices — list (S2)

Purpose. The roster of every device (clients **and** connectors — one `devices` table, `kind` discriminates [CP §2.2, REG §2]). Filter, sort, search, and drill in. Live presence from Redis TTL keys via the stream [REG §5.3, TEL §5].

Layout (data table + filter bar).

```
┌──────────────────────────────────────────────────────────────────
│ Devices (142)        [ search name / overlay-ip ]   [+ Enroll
device ▾]    │
│ Filter: ◉ all  ○ clients  ○ connectors    Status: ○ online ○
offline ○ all │
├───┬──────────────┬─────────┬────────────────────┬──────────┬────
│ ● │ NAME         │ KIND    │ OVERLAY IP         │ OS       │
LAST SEEN     │
├───┼──────────────┼─────────┼────────────────────┼──────────┼────
│ ● │ alice-mbp    │ client  │ fd7a:helix:1::2    │ macos    │
online        │ ← green presence dot
│ ● │ warehouse-gw │ connect.│ fd7a:helix:1::5    │ linux    │
online        │
│ ○ │ bob-pixel    │ client  │ fd7a:helix:1::9    │ android  │ 3 h
ago       │ ← grey (offline)
│ ⊘ │ old-laptop   │ client  │ fd7a:helix:1::4    │ windows  │
revoked       │ ← red (revoked)
│ … │              │         │                    │
│               │
├───┴──────────────┴─────────┴────────────────────┴──────────┴────
│  118 online · 23 offline · 1 revoked           ‹ 1 2 3 … ›
rows: 25 ▾     │
└──────────────────────────────────────────────────────────────────
```

Interactions. Click a row → device detail (S2d). The `[+ Enroll device ▾]` split-button mints a single-use enroll token [CP §9.2] and shows it as text + QR (operator hands it to the device); the QR/token is **never** logged (§11.4.10). Column sort; multi-facet filter; debounced search over name + overlay IP. Presence dot uses the connection-state palette: green online / grey offline / red revoked — **always paired with the text status** (CI5).

States. *Loading* — table skeleton rows. *Empty* — "No devices yet — enroll your first connector or client" with the enroll CTA. *Error* — a banner ("couldn't load devices — retry") above the last-known rows (stale, marked). *Live* — `device.online`/`.offline`/`.revoked`/`.enrolled` events mutate rows in place; a newly enrolled device animates in (`motion.stateXfade` [DT §6.4]).

5.2 Device — detail (S2d)

Purpose. Everything known about one device (identity only, never traffic): identity, overlay IP, presence, current transport/RTT (presence/health only, C3), certificate state, group memberships, and the **revoke** action. For a connector it also surfaces its advertised prefixes (links to IPAM).

```
┌──────────────────────────────────────────────────────────────────
│ ‹ Devices   alice-mbp                             ● online
[ Revoke ⊘ ] │
├──────────────────────────────────────────────────────────────────
│ Identity                         │
Certificate                          │
│  Kind      client                │  Serial
9f2c…                      │
│  Owner     alice@acme  (user)    │  Not after 2026-06-26 12:00Z
(23 h)    │
│  OS        macos                 │  Status    valid
✓                    │
│  Overlay   fd7a:helix:1::2       │  Auto-renews over control
channel     │
│  Enrolled  2026-06-20
│                                    │
│  WG pubkey 32B  Yg8…==  (public) │  [ Force re-
issue ]                    │
├──────────────────────────────────┴───────────────────────────────
│ Presence / health (no traffic data —
C3)                                  │
│  Last seen   online (live)     Transport  masque-h3     RTT  23
ms          │
│  Groups
group:admins                                              │
├──────────────────────────────────────────────────────────────────
│ ⓘ HelixVPN stores identity & policy only. There is no
connection,          │
│   destination, or bandwidth record for this device, by
construction.        │
└──────────────────────────────────────────────────────────────────
```

Revoke flow (§11.4.66 confirm). `[ Revoke ⊘ ]` opens a confirmation dialog ("Revoke alice-mbp? Its tunnel is force-closed and its peer is removed from every map within ~1 s [CP §9.3]. This cannot be undone — re-enrollment is required."). On confirm: `POST /v1/devices/:id/revoke` (admin-only [CP §8.1]); the row flips to revoked, an audit `device.revoke` row is written, and the topology graph drops the node live. Revoke is **admin-only**; operator/member see it disabled with a tooltip stating the required role.

States. *Loading* — header skeleton + panels shimmer. *Error* — per-panel inline error (cert panel can fail independently of identity). *Revoked device* — the whole header tints with the danger/revoked treatment and the revoke button becomes a disabled "Revoked" label; re-enroll guidance is shown.

A11y / light-dark. The connector "advertised prefixes" sub-panel (for connectors) links to IPAM. Panels are `surface.raised`; "valid ✓" is `feedback.success`, an expiring-soon cert (< 1 h) is `feedback.warning`. The privacy note is `text.secondary`, present on **every** device surface so the no-logging guarantee is never out of sight.

---

6. Users & SSO (OIDC config, roles)

Purpose. Manage the human roster and how they authenticate. Two halves: **Users & roles** (the `users` table — `admin` > `operator` > `member` [CP §2.2/§8.1]) and **SSO configuration** (the OIDC IdP the tenant trusts — Keycloak/Authentik [CP §9.1]). Admin-only.

```
┌──────────────────────────────────────────────────────────────────
│ Users & SSO                                          [+ Invite
user ]      │
├──────────────────────────── Users
───────────────────────────────────────┤
│ NAME / EMAIL          IDENTITY      ROLE          DEVICES
STATUS          │
│ alice@acme            OIDC          admin   ▾     3
active          │
│ bob@acme              OIDC          operator ▾    2
active          │
│ carol@ext             OIDC          member  ▾     1
active          │
│ (device-token user)   anonymous     member        1         no
PII (C6)     │ ← anon user row
├──────────────────────────── SSO (OIDC)
────────────────────────────────────┤
│  Provider     Keycloak                                 Status  ●
connected     │
│  Issuer URL   https://id.acme.example/realms/
acme                            │
│  Client ID    helix-
console                                                 │
│  Client secret  •••••••••  [ rotate ]              (never
displayed/logged)│
│  Group claim  groups   → maps IdP groups to Helix
groups                     │
│  [ Test connection ]
[ Save ]                        │
└──────────────────────────────────────────────────────────────────
```

Interactions. Inline **role editor** per user (a dropdown that writes the new `role`; an admin cannot demote the last remaining admin — guarded with an explanatory disable). **Invite** mints an OIDC-bound invitation or, in privacy mode, points to the anonymous device-token path. **SSO config** edits the issuer / client-id / claim mapping; the **client secret is masked, never rendered or logged** (§11.4.10) and only ever rotated. `[ Test connection ]` performs a live OIDC discovery probe and reports the honest result.

States. *Loading* — roster skeleton. *Empty* — a single bootstrap admin + "no SSO configured yet — using local/bootstrap auth" guidance. *Error* — a failed SSO test shows the precise discovery/JWKS error (`feedback.error`), never a vague failure. The **anonymous user row** is rendered with a "no PII" badge to make the privacy posture (C6) visible, not hidden.

A11y / light-dark. Role dropdowns are keyboard-operable; the secret field has an `aria-label` and a copy-suppressed value. Status dots ("connected") pair with text. Light: `surface.default`/`surface.raised`; the rotate/secret controls use `feedback.warning` emphasis to signal sensitivity.

---

7. Policy editor (ACL rules, compile / preview, fail-closed)

Purpose. The Console's highest-stakes screen — author the declarative ACL document (`policies.spec` jsonb [CP §7.1, POL §2]), **dry-run compile** it, inspect the **effect-diff** vs the active version, and **activate** (with instant rollback). This is where "1 user → N networks" becomes compiled per-node visibility (default-deny, fail-closed [POL §0.1]).

Layout (three-pane: editor │ effect-diff │ version history).

```
┌──────────────────────────────────────────────────────────────────────────
│ Policy editor         active: v8       editing: v9 (draft)   ⊘
not saved │
├───────────────────────────┬─────────────────────────────┬──────────────
│ SPEC (declarative ACL)   │ EFFECT PREVIEW (dry-run)    │
VERSIONS         │
│ ┌──────────────────────┐ │  Compile: ✓ ok (0.04 s)     │  v9
draft  ●      │
│ │ groups:              │ │  ───────────────────────    │  v8
active ✓      │
│ │   group:admins: […]  │ │  + admins → office-lan:*     │
v7       ↺      │
│ │ hosts:               │ │  + admins → warehouse-cams   │
v6       ↺      │
│ │   warehouse-cams:    │ │  − contractors → office-lan  │
v5       ↺      │
│ │     10.10.0.0/24     │ │  (removed from prior draft)  │
…               │
│ │ acls:                │ │  ───────────────────────    │  [ diff
v8↔v9 ]   │
│ │  - action: accept    │ │  Visible-peer changes:
│                        │
│ │    src: [group:admins│ │   alice-mbp  +2  −0
│                        │
│ │    dst: ["*:*"]      │ │   carol-lap  +0  −1
│                        │
│ │  - action: accept    │ │  ───────────────────────
│                        │
│ │    src:[group:contr.]│ │  ⚠ advisory: warehouse-cams
```

```
│                     │
│ │    dst:["warehouse-  │ │     overlaps office-lan CIDR
│                     │
│ │      cams:554,80"]   │ │     → resolved by 4via6
│                     │
│ │ exitNodes:[group:adm]│ │
│                     │
│ └──────────────────────┘ │  [ Validate ]  [ Activate
v9 ]                   │
├──────────────────────────┴──────────────────────────┴──────────
│ ✗ ERROR (fail-closed): host "cam-net" dst CIDR not covered by
any          │ ← blocking error state
│   advertised prefix → fix in IPAM, or correct the host.   [ go
to IPAM ]   │
└──────────────────────────────────────────────────────────────
```

Components. `PolicyEditor` [CLIENT §7.2] — a structured text/JSONC editor with syntax highlighting of the ACL grammar [POL §2.3]; the **effect-diff** pane shows the dry-run `CompiledPolicy` delta vs the active version (added/removed edges, per-device visible-peer +/− counts [POL §4.3]); a **version history** rail listing every `policies.version` with the active one marked and each prior one **one-click re-activatable** (instant rollback [POL §6, CP §7.4]).

Fail-closed behaviour (the load-bearing UX). Activation is **disabled** until a clean dry-run [POL §5]. Blocking errors (unknown group/host; a `dst` CIDR not covered by any `advertised_prefixes`; a rule granting a *revoked* device; an `exitNodes` entry resolving to a connector) render in the bottom error bar with the **exact** offending token and an actionable link (e.g. "→ go to IPAM") — never a generic "invalid". Advisory findings (overlapping-CIDR `route.conflict`) appear as warnings in the preview and **do not block** activation [POL §5.1, REG §6.3].

Activation flow. `[ Activate v9 ]` → confirm dialog summarising the effect-diff ("activate v9: 2 devices gain access, 1 loses access; convergence ~<1 s, no restart [CP §10.1]"). On confirm → `POST /v1/policies/{v}/activate`; the active marker moves to v9, an audit `policy.activate` row is written, and the topology + device maps reconcile live. A bad activation is rolled back by re-activating an older version from the rail (instant, no recompile [POL §6.3]).

States. *Loading* — editor + panes skeleton; the active version loads first. *Empty* — a brand-new tenant shows a **starter template** (default-deny + a commented example) rather than a blank document. *Compiling* — the preview shows a "compiling…" state with the elapsed timer; a clean result flips to the diff. *Error* — blocking compile errors (red bar, activate disabled); the editor highlights the offending line. *Dirty* — an unsaved draft shows "⊘ not saved" and warns on navigate-away.

A11y / light-dark. The editor is fully keyboard-navigable with a screen-reader description of compile status (announced, not colour-only). Added edges are `feedback.success` green, removed are `feedback.error` red, advisory is `feedback.warning` amber — each with a +/−/⚠ glyph so the diff is legible without colour (CI5). Light: editor on `surface.sunken` (code well), panes on `surface.raised`; dark mirrors with the same roles.

---

8. Topology (the coordinator graph)

Purpose. Visualise the tenant's overlay as the **coordinator's in-memory topology graph** [COORD §1] — devices, connectors, advertised prefixes, group membership, and *who-can-reach-whom* edges as compiled by the active policy (need-to-know, C4). It is the spatial companion to the tabular Devices/Policy views and updates live.

Layout (graph canvas + inspector).

```
┌──────────────────────────────────────────────────────────────────
│ Topology — acme-corp        [ layout: force ▾ ] [ filter: group
▾ ] ◉ live │
├─────────────────────────────────────────────────────┬────────────
│                                                     │
INSPECTOR                  │
│        (alice-mbp)──────┐                            │
───────────              │
│            │           ▼                            │  Node
warehouse-gw  │
│            │       ╔══════════╗    advertises         │  Kind
connector     │
│            ▼       ║ GATEWAY  ║◄──────(warehouse-gw)│  Site
warehouse     │
│      (bob-pixel)   ╚══════════╝      10.10.0.0/24  │
Prefixes               │
│            ▲           ▲                            │
10.10.0.0/24 ✓      │
│            │           │                            │  Reachable
by          │
│        (carol-lap)─────┘                            │
group:admins        │
│                                                     │  Presence
● online  │
│   ● online  ○ offline  ⊘ revoked   ── reach edge   │  [ open in
IPAM ]    │
└─────────────────────────────────────────────────────┴────────────
```

Components. A graph canvas (`UNVERIFIED`: the concrete graph-render library is pinned by the component spec — stated here as a force/hierarchical layout choice, not a fixed lib). **Nodes** = devices/connectors/gateway, coloured by presence (connection-state palette, green/grey/red, **with a glyph + label**, never colour-only). **Edges** = reach relationships from the compiled policy (`visible[d]` [POL §4.1]). An **inspector** pane shows the selected node's identity, prefixes, who-can-reach-it, presence, and deep-links (connector → IPAM, device → device detail).

Interactions. Click a node → inspector + cross-links; pan/zoom; filter by group or kind; layout toggle. Live: `device.online/offline/revoked`, `connector.prefixes.changed`, and `policy.compiled` events animate the graph (node colour change, edge add/remove) without refresh [COORD §4.2].

States. *Loading* — graph skeleton (placeholder node ring) while the snapshot hydrates. *Empty* — a 1-node graph (just the gateway) with "enroll a connector or client to grow your network". *Error* — if the snapshot fails, an inline retry over the last-known graph (marked stale). *Large-graph* — beyond a node threshold the view clusters by group/site and offers search-to-focus (so the canvas stays legible — no overlap, §11.4.162; nodes are spaced and labels never overlay each other).

A11y / light-dark. A **table fallback** of the same graph data is available (the graph is reinforcement, not the only way to read topology — keyboard/SR users get the tabular Devices + a reach matrix). Canvas background is `background.canvas`; edges use `border.strong`; the danger/revoked node uses the red state colour at z-top so it is never occluded [COLOR §3.2/§5].

---

9. Audit log (control-action audit — no traffic logs)

Purpose. The tamper-evident record of **control actions** — device enroll/revoke, policy create/activate, prefix changes, user/role changes, cert issue/revoke (`audit_events`, closed `AuditAction` vocabulary [TEL §4.2]). The screen's defining feature is what it **cannot** show: there is **no** connection, destination, flow, or bandwidth record — by construction (C3 [CP §0.1/§2.4]).

```
┌──────────────────────────────────────────────────────────────────────
│ Audit                          [ action ▾ ] [ actor ▾ ]
[ 2026-06-25 range ] │
│ ⓘ This is a CONTROL-action log. HelixVPN keeps NO traffic,
connection,    │ ← the no-traffic banner
│   destination, or bandwidth records — that data does not exist
(C3).      │
├──────────────┬──────────┬────────────────────┬────────────────────
│ TIME (UTC)   │ ACTOR    │ ACTION             │ TARGET /
META              │
├──────────────┼──────────┼────────────────────┼────────────────────
│ 12:04:31     │ alice    │ device.revoke      │ dev-31 (old-
laptop)         │
│ 12:03:10     │ system   │ policy.activate    │ v8  → see Policy
editor       │
│ 12:01:55     │ bob      │ device.enroll      │ connector
warehouse-gw       │
│ 11:58:02     │ carol    │ prefixes.change    │ +10.20.0.0/24 on
conn-B      │
│ 11:40:18     │ alice    │ user.role.change   │ carol
member→operator         │
│ …            │          │
│                              │
├──────────────┴──────────┴────────────────────┴────────────────────
│              ‹ newer   older ›    export: CSV ▾  (control
actions only)    │
└──────────────────────────────────────────────────────────────────────
```

Components. A reverse-chronological feed (indexed `audit_events (tenant_id, ts DESC)` [CP §2.2]); **filters** by closed `action` vocabulary + actor + time range; deep-links from a row's target to the relevant screen (a `policy.activate` row → that version in S4; a `device.revoke` row → S2d). Export is **CSV of control actions only** — the export pipe shares the no-traffic guarantee.

The privacy banner is load-bearing. It is always visible at the top of S7 and restated near the export control: the Console is *incapable* of producing a traffic log because the schema holds none (the §2.4 schema-lint is the runtime proof [CP §2.4]). This turns "we don't log" from a promise into a visible product property.

States. *Loading* — feed skeleton. *Empty* — "no control actions in this range" (a legitimate empty, not an error). *Error* — banner + retry over last-known rows. *Read authz* — audit **read** is admin-gated [TEL §4.5]; operator/member see a "audit is admin-only" state, not a partial leak.

A11y / light-dark. The feed is a semantic table; rows are keyboard-navigable; the privacy banner uses `feedback.info` (blue) so it reads as informational, not alarming. Light/dark via standard surface/text roles.

---

10. IPAM / prefixes

Purpose. The address plane: the tenant's overlay **ULA /48** pool [IPAM §1, CP §3] and the **advertised prefixes** each connector exposes [REG §6], including the **overlapping-CIDR conflict** surface and the `4via6` site-disambiguation that resolves it (D4). Operator+.

```
┌──────────────────────────────────────────────────────────────────
│ IPAM / prefixes — acme-
corp                                                    │
├────────────────────────── Overlay pool
─────────────────────────────────────┤
│  ULA /48     fd7a:helix:a1b2::/48        allocated hosts
144             │
│  Gateway     fd7a:helix:a1b2::1          next
host       ::92            │
│  ⓘ /48 capacity is astronomically large; hosts are never
recycled (IPAM).  │
├───────────────────── Advertised prefixes (by connector)
────────────────────┤
│  CONNECTOR       SITE-ID   CIDR            ENABLED
STATUS              │
│  warehouse-gw    site-1    10.10.0.0/24    ✓
ok                  │
│  office-gw       site-2    192.168.50.0/24 ✓
ok                  │
│  branch-gw       site-3    192.168.50.0/24 ✓         ⚠ overlaps
office-gw  │ ← conflict row
├───────────────────────────── Conflicts
─────────────────────────────────────┤
│  ⚠ 192.168.50.0/24 advertised by office-gw (site-2) AND branch-
gw (site-3) │
│    Advisory — not blocking. Resolved by 4via6 site
disambiguation:         │
│    office  → fd7a:helix:a1b2:0002::/96   branch →
…:0003::/96               │
│    [ accept 4via6 mapping ]   [ choose authoritative
```

```
connector ]               │
└──────────────────────────────────────────────────────────────────
```

Components. An **overlay-pool card** (ULA prefix, gateway address, next host, allocated count [IPAM §1.2]); an **advertised-prefix table** grouped by connector with site-id, CIDR, enable toggle, and per-row status; a **conflicts panel** listing overlapping-CIDR `route.conflict.detected` items [REG §6.2] with the **4via6 resolution** spelled out (the per-site `/96` mapping the control plane ships in the network map [IPAM §4.3]) and the two operator choices (accept 4via6 / designate authoritative).

Interactions. Toggle a prefix `enabled`; edit a connector's CIDRs (writes the same path as the connector-side advertise, [REG §6.1] — links to the Connector app for the source). Resolve a conflict via the offered choices (§11.4.66 — never auto-resolved silently). Cross-link: the Policy editor's "host CIDR not advertised" error lands here (S4 ? S6).

States. *Loading* — pool/table skeleton. *Empty* — "no connectors advertising prefixes yet". *Conflict present* — the conflict panel is highlighted (`feedback.warning`) and a count badges the nav item; it is **advisory** (never blocks policy activation [POL §5.1]). *Error* — per-section inline retry.

A11y / light-dark. Conflict rows carry a ⚠ glyph + text (not colour-only); the 4via6 mapping is shown as monospace addresses (`font.primitive.family.mono` [DT §4]). Light/dark via surface/text/feedback roles; the conflict highlight is amber in both themes.

---

11. Telemetry / health (counters, convergence & event-lag SLOs)

Purpose. The operator's window into whether the control plane is **meeting its measured SLOs** [TEL §3, CP §10.2]: convergence p99 (event → delta-on-wire < 1 s), revoke latency (< 1 s), event lag, stream pending depth, and runtime health — all from Prometheus series the telemetry service emits [TEL §3.2/§3.3]. This is the anti-bluff "is it actually fast?" screen.

```
┌──────────────────────────────────────────────────────────────────
│ Telemetry / health — acme-corp                          window:
1h ▾        │
├───────────────┬───────────────┬───────────────┬──────────────────
│ CONVERGENCE   │ REVOKE        │ EVENT LAG     │ STREAM
DEPTH                │
│ p99 0.42 s    │ p99 0.31 s    │ 0.08 s        │ events:policy
0           │
│ target <1s ✓  │ target <1s ✓  │ healthy ✓     │ events:routes
2           │
│ ▲ green       │ ▲ green       │ ▲ green       │ events:presence
0           │
├───────────────┴───────────────┴───────────────┴──────────────────
│  helix_reconcile_seconds (event → delta) — last
1h                            │
│  1s┤ — — — — — — — — — — — — — — — — — — — — — — — (SLO line) —
```

```
- - - - -   │
│    │
┌┐                                                        │
│ .5s┤    ┌──┐  ┌───┐  ┌──┘└──┐   ┌───┐       (p99 histogram
band)                  │
│    │ ──┘  └─┘    └─┘      └──┘
└────                                           │
│  0
└──────────────────────────────────────────────────────────────────
│
├──────────────────────────────────────────────────────────────────
│
ALERTS
│
│   ● none firing      (SLO breach → alert rules live in deploy/,
TEL §3.4)   │
│
RUNTIME
│
│   open streams 124   coordinator RSS 412 MB (24h slope ≈ 0)  DLQ
0          │
└──────────────────────────────────────────────────────────────────
```

Components. Four **SLO gauges** (convergence p99, revoke p99, event lag, stream depth) each green/amber/red against its target [TEL §3.4]; a **convergence histogram** with the SLO line drawn at 1 s so a breach is visually obvious; an **alerts** list (the *series* are emitted here, the *alert rules* live in `deploy/` [TEL §3.4] — the screen states this honestly); a **runtime** strip (open stream count, coordinator RSS with the 24 h ≈0 slope soak signal [CP §10.2], DLQ depth `helix_events_dlq_total` [CP §5.4]).

Honest-SLO logic (§11.4.6). Every gauge shows the *measured* number and its target; " ✓ " appears only when measured < target. A breach turns the gauge red and surfaces in the alerts list — the screen never shows "healthy" without the figure that justifies it. This is the visible face of the anti-bluff < 1 s promise.

States. *Loading* — gauge/chart skeleton. *Empty* — a just-bootstrapped tenant with no traffic yet shows "no data in window — generate activity or widen the window". *Error* — if the metrics scrape fails, the screen says so (it does **not** paint green on absence of data — absence ≠ healthy). *Breach* — red gauge + alert row + a link to the offending series.

A11y / light-dark. Gauges pair colour with a value + ✓/⚠/✗ glyph; the SLO line is labelled. Charts have an accessible data-table fallback. Light/dark via standard roles; the SLO line uses `feedback.info`, breaches `feedback.error`.

---

12. Billing / subscription

Purpose. Manage the tenant's plan, seat/device usage against plan limits, and invoices. This is a **commercial** surface layered over the self-hostable core — its data source (the billing provider) is not specified by the control-plane docs.

```
┌──────────────────────────────────────────────────────────────────
│ Billing — acme-
corp                                                        │
├───────────────────────── Plan ─────────────────────┬──────────
Usage ────────┤
│  Plan      Team                                    │  Devices
142 / 250    │
│  Renews    2026-07-20                               │  Seats
18 / 25      │
│  Status    ● active                                 │  Connectors
7 / 10     │
│  [ Change plan ]   [ Manage payment method ]        │  ▓▓▓▓▓▓░░░░
57%        │
├───────────────────────── Invoices
──────────────────────────────────────────┤
│  DATE        AMOUNT     STATUS
INVOICE                                  │
│  2026-06-20  $240.00    paid         [ download
PDF ]                       │
│  2026-05-20  $240.00    paid         [ download
PDF ]                       │
│
…
│
└──────────────────────────────────────────────────────────────────
```

Components. A **plan card** (tier, renewal, status, change/payment actions); a **usage panel** (devices/seats/connectors against plan limits with a progress bar); an **invoice table** (date/amount/status/download). Approaching a limit shows a `feedback.warning` nudge; over-limit a `feedback.error` block on new enrollment.

> **D-CON-BILL-1 `UNVERIFIED`.** The billing provider + its API are **not** specified by the control-plane docs [CP] — this screen's data source is a Phase-2 commercial decision (§11.4.66). The layout/states here are the *intended* shape; the provider binding is `UNVERIFIED` until that decision lands. The self-hostable MVP may ship the Console with Billing **hidden** behind a capability flag (no commercial backend) — also `UNVERIFIED`.

States. *Loading* — skeleton. *Empty / self-hosted* — when no billing backend is configured, the screen shows "self-hosted — no subscription" rather than a broken integration. *Error* — provider errors shown honestly with retry.

A11y / light-dark. Usage bars carry a numeric label; status dots pair with text. Standard surface/text/feedback roles, light + dark.

---

13. Settings

Purpose. Tenant-level configuration + per-admin preferences + the danger zone. Sectioned, with role-gated controls.

```
┌──────────────────────────────────────────────────────────────────
│
```

```
Settings
│
├──────────────────────────────────────────────────────────────────
│ ▸ Tenant       name, control-plane URL, default policy
template            │
│ ▸ Appearance   theme  ◐ system ○ light ○ dark       (per-
admin)            │
│ ▸ Session      signed-in devices, sign-out-everywhere, session
TTL          │
│ ▸ Transport    default escalation ladder order (plain-
udp→lwo→masque-h3)    │
│ ▸ Notifications which control events page the live rail / send
alerts        │
│ ▸ Danger zone  rotate tenant CA · delete tenant   (admin-only,
confirmed)   │
└──────────────────────────────────────────────────────────────────
```

Components. Collapsible sections. **Appearance** is the per-admin theme toggle (system/light/dark — both themes first-class [CLIENT §7.1]). **Transport** edits the default `TransportPolicy.order` ladder the control plane ships to agents [CP §4 `TransportPolicy`]. **Danger zone** holds irreversible, admin-only, double-confirmed actions (rotate the tenant CA key [CP §9.3] — the one true secret; delete tenant) each gated per §11.4.66 with a typed-confirmation dialog.

States. *Loading* — section skeletons. *Error* — per-section inline. *Role- gated* — operator/member see read-only or hidden danger-zone controls with the required-role tooltip.

A11y / light-dark. Every toggle is keyboard-operable and labelled; the theme toggle previews live. Danger-zone actions use `feedback.error` emphasis and require typing the tenant name to confirm. Standard roles, light + dark.

---

14. Cross-screen states, a11y & light/dark contract

These hold for **every** Console screen (so each section above states only its specifics):

Concern	Contract	Source
Loading	skeleton placeholders (shimmer), never a bare spinner on a blank screen; the most-important data loads first	original UX
Empty	a purposeful empty state with the next action (enroll, configure, invite) — never an ambiguous blank	original UX
Error	the **honest** cause (provider/JWKS/scrape error), an inline retry, and last-known data marked stale — never a generic "something went wrong"	§11.4.6
Stale/live	the live-stream chip goes amber on WS drop and lists show "may be stale —	CI2 [CLIENT §0.1]

Concern	Contract	Source
	reconnecting"; the UI never paints fresh data it cannot prove is fresh	
Confirm	every destructive control action (revoke, deactivate, rotate CA, delete) is double-confirmed via the interactive mechanism	§11.4.66
Colour ≠ only signal	every state dot/badge pairs colour with a glyph + text label	CI5 [CLIENT §0.1], [COLOR §0]
No overlap / no label-overlay	spacing tokens keep ≥ `space.scale.2` (8 px) between coloured regions; overlays use `overlay.scrim`, never a translucent brand wash; danger sits at z-top	§11.4.162, [COLOR §5]
Light + dark	every screen renders in both, all surface/text/feedback roles AA-proven both themes; the violet accent is the only per-app difference	§11.4.162, [COLOR §2/§4/§6]
Keyboard / SR	full keyboard nav (the Console is desktop-first); ⌘K command palette; data-heavy visuals (graph, charts) have table fallbacks	CI5, [CLIENT §7.3]
No traffic data	no Console screen can show connection/flow/bandwidth records; the Audit screen states this is by construction	C3 [CP §0.1/§2.4]

The violet light/dark accent (the Console's only brand difference).

Role	Light	Dark	Proof [COLOR §6]
`action.primary` / accent	`violet.700` `#6D28D9`	`violet.400` `#A78BFA`	white-on-`violet.600` **5.70**; text **7.10** light / **6.47** dark — AA both

Everything else — surfaces, text, borders, **connection-state**, **feedback** — is the identical shared semantic set used by Client and Connector (D-COLOR-1); the Console never recolours a safety signal.

---

15. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-CON-1	Console is desktop-first (`lg`/`xl` home turf) but renders the full responsive ladder down to `compact`; the graph + policy editor + heavy tables deferred-load so they never bloat a small layout.	decided
D-CON-2	Live data via `GET /v1/stream` WS/SSE folded into Riverpod; a visible stream-health chip + "may be stale" honesty when it drops.	decided [CLIENT §8.3]
D-CON-3	The Audit screen makes the no-traffic-log guarantee a *visible product property* (banner +	decided [CP §2.4]

id	Decision / contract	Status
	control-actions-only export), backed by the §2.4 schema-lint runtime proof.	
D-CON-4	Policy activation is fail-closed: disabled until a clean dry-run; blocking errors name the exact token + an actionable link; advisory CIDR conflicts never block.	decided [POL §5]
C-CON-A (consumes)	All control-plane data + payloads owned by [CP §8] + the V3 `svc-*.md` docs; this doc owns only layout/state/interaction over them.	contract
C-CON-B (consumes)	Colour roles + violet accent owned by [COLOR]; token scales/breakpoints by [DT]; signature components by [`component-library.md`].	contract
C-CON-C (provides)	This screen set + per-screen states are the spec the visual-regression + golden suite asserts against (light+dark, every breakpoint, §11.4.162).	contract
U-CON-1 `UNVERIFIED`	Billing provider + API (S12) — a Phase-2 commercial decision (§11.4.66); the MVP may ship Billing hidden behind a capability flag.	open
U-CON-2 `UNVERIFIED`	The concrete topology graph-render library (S8) is pinned by the `component-library.md` spec; layout choice (force/hierarchical) stated here, library `UNVERIFIED`.	open
U-CON-3 `UNVERIFIED`	The exact `GET /v1/stream` event field set the live rail/dashboard digest renders is owned by [CP §8]; the digest's column choice is stated here, the wire shape `UNVERIFIED` against a frozen WS contract test.	open

---

Sources verified

- **Console flavor (admin, no `core_ffi`, Web-only build), capability model, `AdaptiveScaffold`, responsive law, ⌘K palette, live-WS folding, theme-per-role, signature components (`PolicyEditor`/`NetworkTile`/…), a11y (colour ≠ only signal)** — `final/03-client-core-and-ui.md` §6 (flavors/capabilities), §7 (design system + responsive + a11y), §8.3 (Console live data), §0.1 (CI2/CI3/CI5) (read 2026-06-25).
- **Control-plane data every screen surfaces** — `final/02-control-plane.md`: §2.2 (devices/users/policies/audit DDL), §3 (IPAM ULA /48 + 4via6, D4), §4 (`Coordinator` proto + `TransportPolicy`), §7 (policy model/compiler, fail-closed, activate/rollback), §8 (REST + WS/SSE + RBAC), §9 (identity/enrollment/PKI/revoke < 1 s), §10.2 (measured SLOs), §0.1 C3 (no-logging by construction) + §2.4 (schema-lint) (read 2026-06-25).
- **Per-service screen data** — `final/v03-control-plane/`: `svc-registry.md` §2/§5/§6 (device model, presence, prefix conflict detection); `svc-policy.md` §2/§4/§5/§6 (ACL spec, compile, validation, activation/rollback, effect data); `svc-coordinator.md` §1/§4 (topology graph + event reactions); `svc-telemetry.md` §3/§4/§5 (Prometheus series, audit sink + closed `AuditAction`

vocab, presence); `svc-ipam.md` §1/§4 (ULA address plan, 4via6 derivation); `svc-identity.md` (OIDC/anonymous identity modes) (read 2026-06-25).

- **Colour roles, the Console violet accent + its AA contrast proofs, the no-overlap / no-label-overlay colour rule, light+dark mandate** — `final/v10-design/color-system.md` §2 (semantic roles), §3 (connection-state palette), §4 (contrast proofs), §5 (no-overlap rule), §6 (per-app accent: Console = violet) (sibling, this wave).
- **Token tiers, spacing/radius/elevation/motion/z/breakpoint scales (the responsive breakpoints + the surfaces/shadows the layouts use)** — `final/v10-design/design-tokens.md` §6 (scales), §1 (tiers) (sibling, this wave).
- **OpenDesign engine that emits the tokens** — `final/v10-design/opendesign-foundation.md` (sibling, this wave; §11.4.162).
- **Signature-component contracts (`AdaptiveScaffold`, `PolicyEditor`, `NetworkTile`, graph canvas) + the Connector/Client counterpart screens** — `final/v10-design/component-library.md`, `…/screens-connector.md`, `…/screens-client.md` (siblings; component-library + client-screens land in this same Volume-10 wave — cross-referenced, not duplicated).
- Items marked `UNVERIFIED` (U-CON-1 billing provider, U-CON-2 graph-render library, U-CON-3 frozen WS field set) are pending their named decision/contract test per §11.4.6 — **not** asserted as fact.
- **Layout, per-screen state machines, interaction flows, wireframes — NO external source needed — original HelixVPN UX design work** (the screen compositions are owned by this document; they render the cited control-plane data through the cited design-system tokens/components).